All About a Configuration Management Tool Called Chef

Explore Chef, the open source tool that streamlines the task of configuring and maintaining a company's servers. Written in Ruby and Erlang, it can easily integrate with cloud-based platforms to automatically provision and configure new machines.

Chef is an open source tool and framework that provides developers and systems administrators a foundation of APIs and libraries, which are used to perform tasks such as configuring and managing IT infrastructures. The concept behind using framework technology is to reduce development time by minimising the overhead of implementing and managing the low-level details that support the development effort. This enables organisations to concentrate on delivery and meeting the customers' requirements.

Chef is a framework for infrastructure development, which provides a supporting structure and package for framing one's infrastructure as code. It provides extensive libraries for building up an infrastructure that can be deployed within a software project. It provides a Ruby-based DSL modelling infrastructure, which enables developers and systems administrators to build and design scalable environments without considering the operating systems. It produces consistent, shareable and reusable components.

Chef provides various tools built on the framework for designing the infrastructure. These tools are based on high level programming languages such as Python and Ruby. Some of the tools are:

1) *Ohai*, a system profiling tool, which is an extendable plug-in to furnish the data that is used to build up the database of each system that is managed by Chef.
2) *Shef*, an interactive debugging console for testing and exploring Chef's behaviour in various conditions. It provides command line access to framework libraries.
3) *Chef-Solo*, a configuration management tool, which allows access to subsets of Chef's features.
4) *Chef Client*, an agent that runs on the system managed by Chef to communicate with the Chef server.
5) *Knife*, a multi-purpose command line tool, which provides system automation, integration, and deployment.

A few features of Chef

- It manages a huge amount of nodes on a single Chef server.
- It maintains a blueprint of the entire infrastructure.
- It can be easily managed using operating systems such as Linux, Windows, FreeBSD and Solaris.
- It integrates with all major cloud service providers.
- High availability.
- Centralised management, i.e., a single Chef server can be used as the centre for deploying the policies.
- It provides a Web-based management console to control the access of objects such as nodes, cookbooks and so on.

How Chef works

Chef converts infrastructure into code. Whatever operations can be done with the code, can be done with the infrastructure as well. Yes, it is possible to keep infrastructure versions, test them and create infrastructure repeatedly and efficiently.

A resource is a definition of an action that can be taken such as the installation of a package, Web server or application server; or managing a configuration file. Chef uses reusable components known as recipes to automate infrastructure. A recipe is a collection of resources that includes packages, templates and so on. A cookbook is a set of recipes such as the installation of the Web server, the database server, and configuration file changes to connect the Web and database servers in different deployment environments. A workstation is the place where the Knife is configured to upload cookbooks to the Chef server. Chef's command-line tool, Knife, is used to interact with the Chef server and upload cookbooks. Attributes provide specific details of a node, such as the IP address and other configuration details.

The Chef server is the central registry or brain of the entire process and can act as a hub. It consists of information about the infrastructure and cookbooks. The latter are used to instruct Chef on how each node must be configured. The Chef server can be either hosted on-premise, on the cloud or be offered by